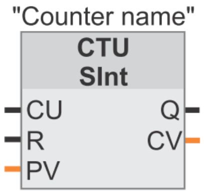
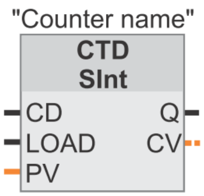
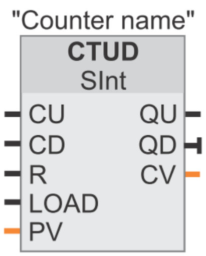
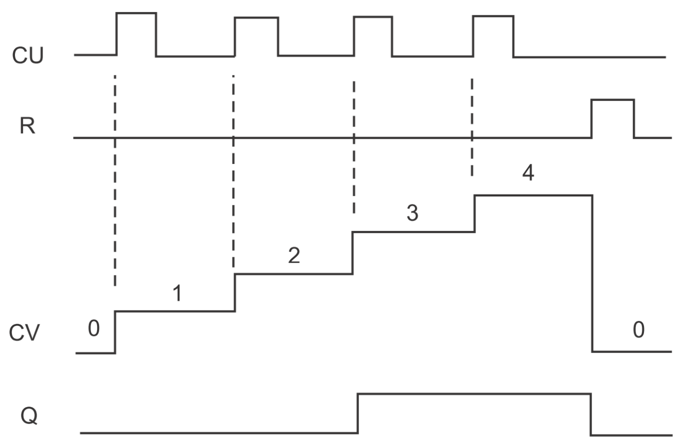
Counters - external pulses







CTU



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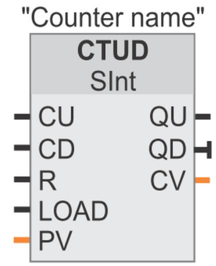

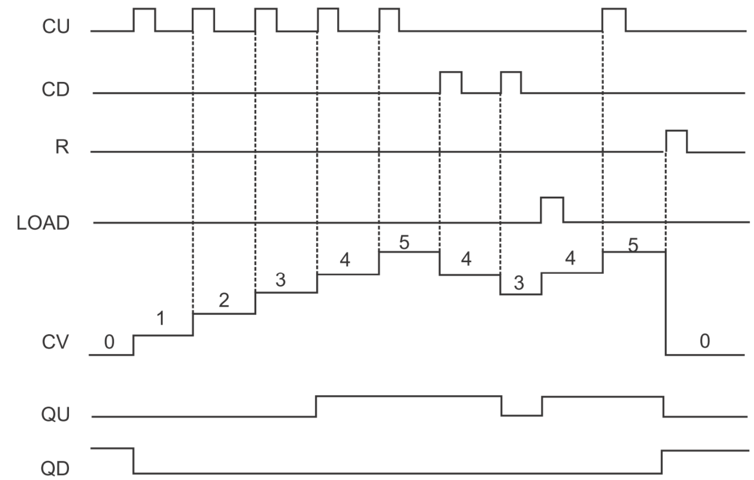

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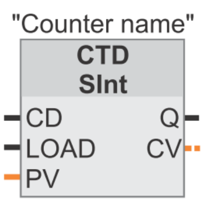

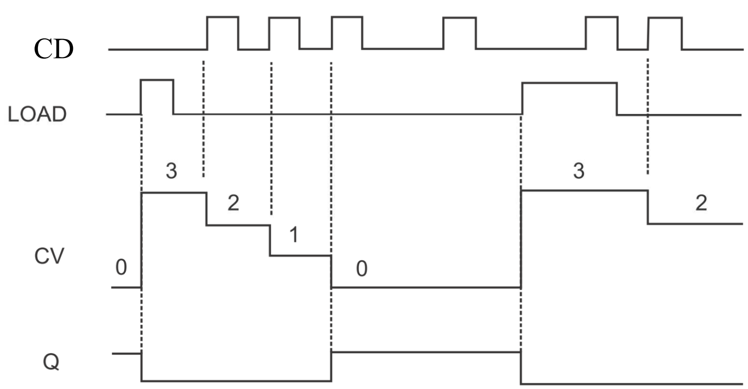

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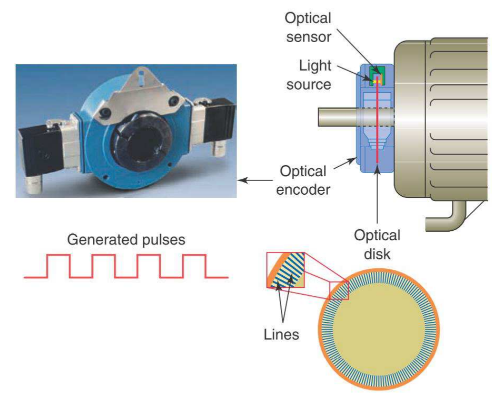
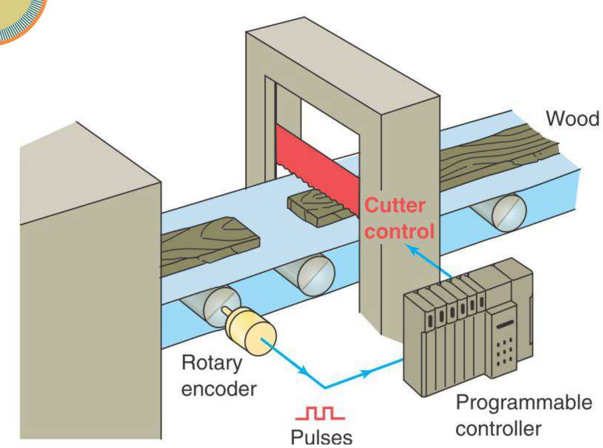
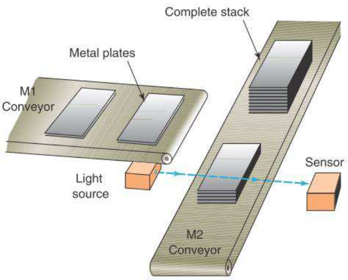
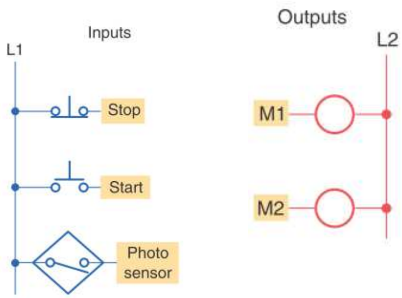


Figure 8-30 Optical incremental encoder.








```
PROGRAM  PLC_PRG
VAR
    //Inputs
    Stop : BOOL ;  // NC
    Start : BOOL ;  //NO
    Sensor : BOOL ;  //NO
    //Outpus
    M1 , M2 , NormalWork :  BOOL ;
    //Constants
    RefNumber :  WORD  :=  4 ;
    TimeM2 :  TIME  :=  T#10S ;
    // Variables & Blocks
    Counter1 :  CTU ;
    CurrentNumber :  WORD ;
    Timer2 :  TON ;
    CurrentTime :  TIME ;
END_VAR
```


Literals - fixed values:

2#11001010

8#377

156

12.0   132.3e-3

Literals of the time description:

t# lub **T#** lub **time#** lub **TIME#**

Units: D days, H hours, M minutes, S seconds and MS milliseconds

The base is 1 ms and the range is from 0 to 2^{32}-1 ms, it means t#49D17H

Example: 4m15s7ms

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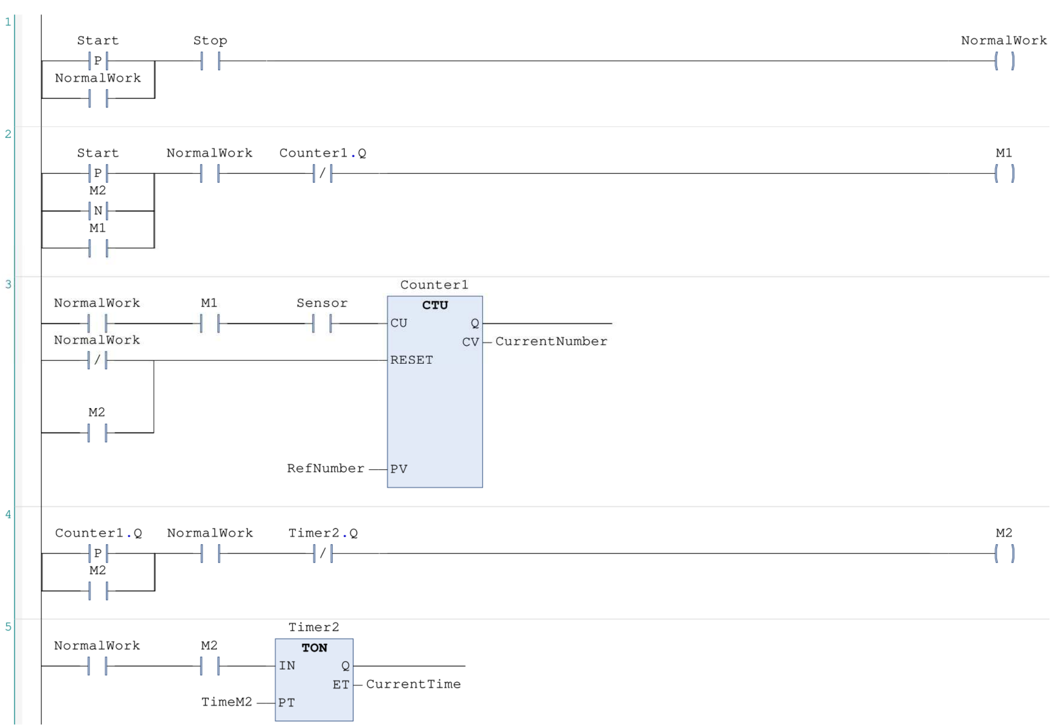

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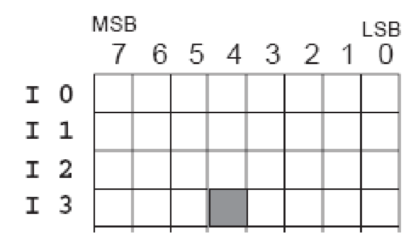
Bit addressing



I 3 . 4

Bit number in byte (calculated from right to left) - digit from 0 to 7

Sequential number of byte in a given address area

Area identifier. The most important of these are:
I - the discrete input area
Q - discrete output area
M - internal memory area

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Obszar pamięci	Opis	Wymuszony	Trwały
I obraz procesu – wejście I_:P (fizyczne wejście)	Skopiowany na początku cyklu programu stan wejść fizycznych	Tak	Nie
	Bezpośredni odczyt wejściowych punktów fizycznych CPU, SB, SM	Nie	Nie
Q obraz procesu – wyjście Q_:P (fizyczne wyjście)	Stan skopiowany na początku cyklu programu do wyjść fizycznych	Tak	Nie
	Bezpośredni zapis do wyjściowych punktów fizycznych CPU, SB, SM	Nie	Nie
M pamięć bitowa	Pamięć sterująca i danych	Nie	Tak
L pamięć chwilowa	Chwilowe dane dla bloku, lokalne dla tego bloku	Nie	Nie
DB blok danych	Pamięć danych, jak również parametrów dla FB	Nie	Tak

Address of bytes and words

M B 100

Initial address of the object

Object length identifier:

B - byte (8 bits)

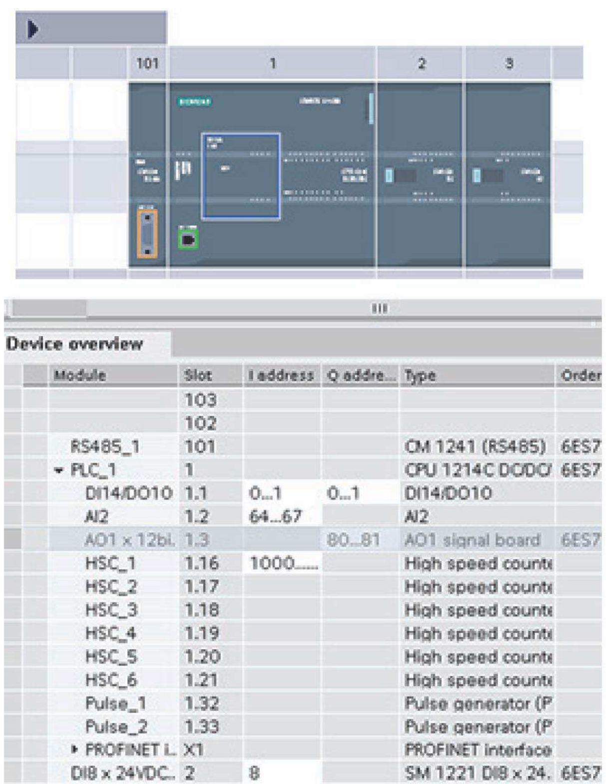
W - word (2 bytes)

D - double word (4 bytes)

Area identifier:

M - internal memory

I, Q - discrete inputs and outputs



Module	Slot	I address	Q addre...	Type	Order
	103				
	102				
RS485_1	101			CM 1241 (RS485)	6ES7
▾ PLC_1	1			CPU 1214C DC/DC/	6ES7
DI14/DO10	1.1	0...1	0...1	DI14/DO10	
AI2	1.2	64...67		AI2	
AO1 x 12bi.	1.3		80...81	AO1 signal board	6ES7
HSC_1	1.16	1000.....		High speed counte	
HSC_2	1.17			High speed counte	
HSC_3	1.18			High speed counte	
HSC_4	1.19			High speed counte	
HSC_5	1.20			High speed counte	
HSC_6	1.21			High speed counte	
Pulse_1	1.32			Pulse generator (P	
Pulse_2	1.33			Pulse generator (P	
▸ PROFINET i.	X1			PROFINET interface	
DI8 x 24VDC..	2	8		SM 1221 DI8 x 24.	6ES7

Address of bytes and words

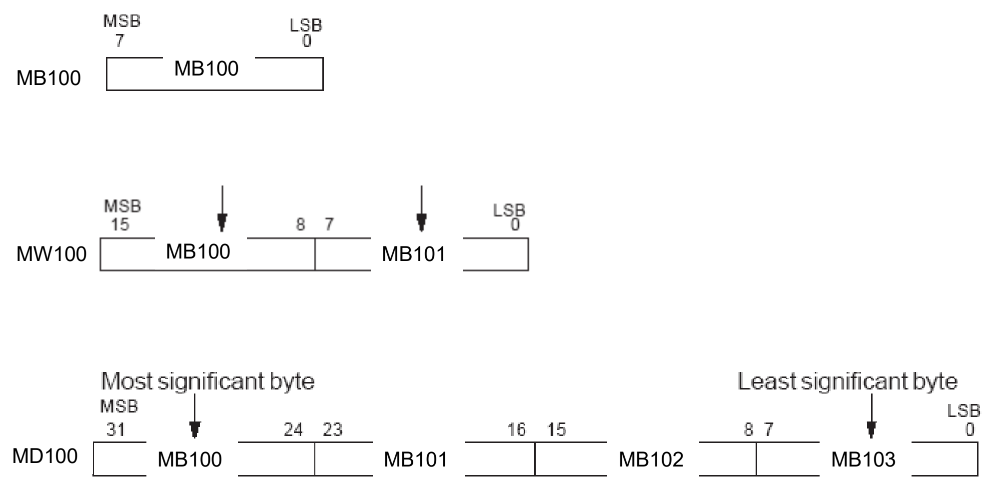


Table 4- 16 Bit and bit sequence data types

Data type	Bit size	Number type	Number range	Constant examples	Address examples
Bool	1	Boolean	FALSE or TRUE	TRUE, 1,	I1.0 Q0.1 M50.7 DB1.DBX2.3 Tag_name
		Binary	0 or 1	0, 2#0	
		Octal	8#0 or 8#1	8#1	
		Hexadecimal	16#0 or 16#1	16#1	
Byte	8	Binary	2#0 to 2#11111111	2#00001111	IB2 MB10 DB1.DBB4 Tag_name
		Unsigned integer	0 to 255	15	
		Octal	8#0 to 8#377	8#17	
		Hexadecimal	B#16#0 to B#16#FF	B#16#F, 16#F	
Word	16	Binary	2#0 to 2#1111111111111111	2#1111000011110000	MW10 DB1.DBW2 Tag_name
		Unsigned integer	0 to 65535	61680	
		Octal	8#0 to 8#177777	8#170360	
		Hexadecimal	W#16#0 to W#16#FFFF, 16#0 to 16#FFFF	W#16#F0F0, 16#F0F0	
DWord	32	Binary	2#0 to 2#111111111111111111111111 11111111	2#111100001111111100 001111	MD10 DB1.DBD8 Tag_name
		Unsigned integer	0 to 4294967295	15793935	
		Octal	8#0 to 8#37777777777	8#74177417	
		Hexadecimal	DW#16#0000_0000 to DW#16#FFFF_FFFF, 16#0000_0000 to 16#FFFF_FFFF	DW#16#F0FF0F, 16#F0FF0F	

Table 4- 17 Integer data types (U = unsigned, S = short, D= double)

Data type	Bit size	Number Range	Constant examples	Address examples
USInt	8	0 to 255	78, 2#01001110	MB0, DB1.DBB4, Tag_name
SInt	8	-128 to 127	+50, 16#50	
UInt	16	0 to 65,535	65295, 0	MW2, DB1.DBW2, Tag_name
Int	16	-32,768 to 32,767	30000, +30000	
UDInt	32	0 to 4,294,967,295	4042322160	MD6, DB1.DBD8, Tag_name
DInt	32	-2,147,483,648 to 2,147,483,647	-2131754992	

Table 4- 18 Floating-point real data types (L=Long)

Data type	Bit size	Number range	Constant Examples	Address examples
Real	32	-3.402823e+38 to -1.175 495e-38, ±0, +1.175 495e-38 to +3.402823e+38	123.456, -3.4, 1.0e-5	MD100, DB1.DBD8, Tag_name
LReal	64	-1.7976931348623158e+308 to -2.2250738585072014e-308, ±0, +2.2250738585072014e-308 to +1.7976931348623158e+308	12345.123456789e40, 1.2E+40	DB_name.var_name Rules: • No direct addressing support • Can be assigned in an OB, FB, or FC block interface table

Table 4- 19 Time and date data types

Data type	Size	Range	Constant Entry Examples
Time	32 bits	T#-24d_20h_31m_23s_648ms to T#24d_20h_31m_23s_647ms Stored as: -2,147,483,648 ms to +2,147,483,647 ms	T#5m_30s T#1d_2h_15m_30s_45ms TIME#10d20h30m20s630ms 500h10000ms 10d20h30m20s630ms
Date	16 bits	D#1990-1-1 to D#2168-12-31	D#2009-12-31 DATE#2009-12-31 2009-12-31
Time_of_Day	32 bits	TOD#0:0:0.0 to TOD#23:59:59.999	TOD#10:20:30.400 TIME_OF_DAY#10:20:30.400 23:10:1
DTL (Date and Time Long)	12 bytes	Min.: DTL#1970-01-01-00:00:00.0 Max.: DTL#2554-12-31-23:59:59.999 999 999	DTL#2008-12-16-20:30:20.250

Table 4- 20 Size and range for DTL

Length (bytes)	Format	Value range	Example of value input
12	Clock and calendar Year-Month-Day:Hour:Minute: Second.Nanoseconds	Min.: DTL#1970-01-01-00:00:00.0 Max.: DTL#2554-12-31-23:59:59.999 999 999	DTL#2008-12-16-20:30:20.250

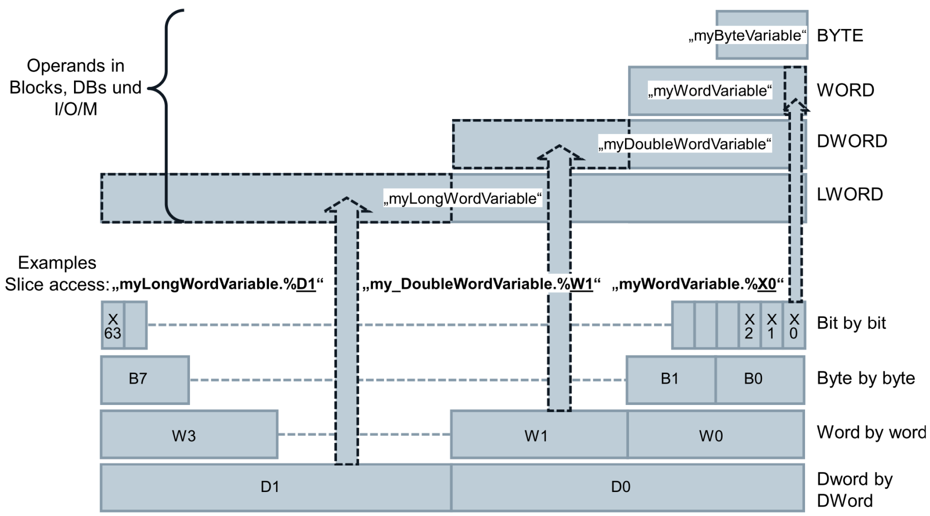
Each component of the DTL contains a different data type and range of values. The data type of a specified value must match the data type of the corresponding components.

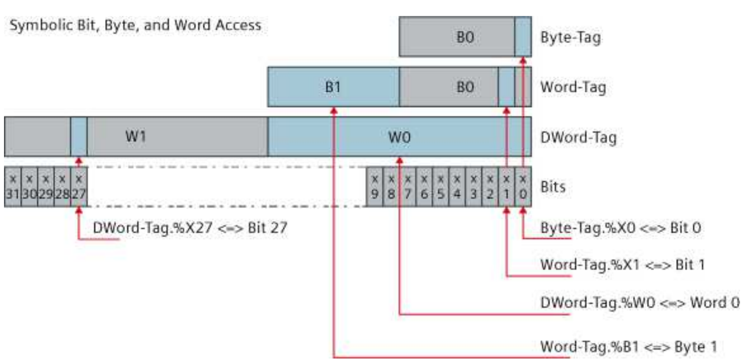
Table 4- 21 Elements of the DTL structure

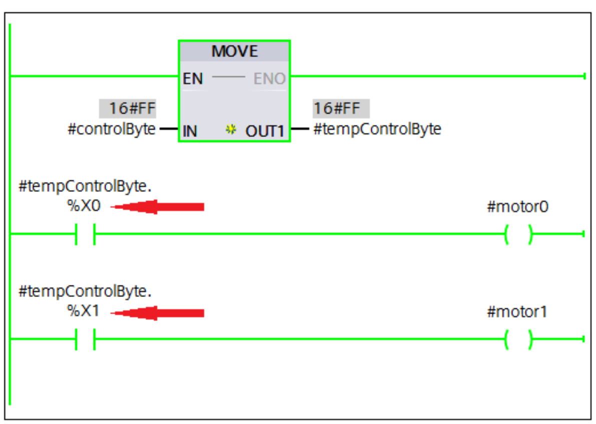
Byte	Component	Data type	Value range
0	Year	UINT	1970 to 2554
1			
2	Month	USINT	1 to 12
3	Day	USINT	1 to 31
4	Weekday [1]	USINT	1(Sunday) to 7(Saturday) [1]
5	Hour	USINT	0 to 23
6	Minute	USINT	0 to 59
7	Second	USINT	0 to 59
8	Nanoseconds	UDINT	0 to 999 999 999
9			
10			
11			

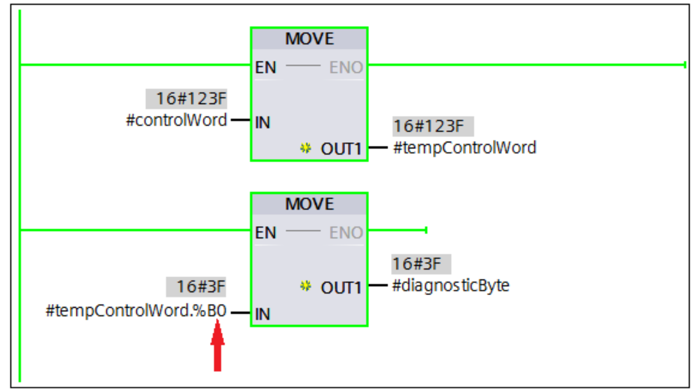
Slice access (Simatic S7 1200/1500)

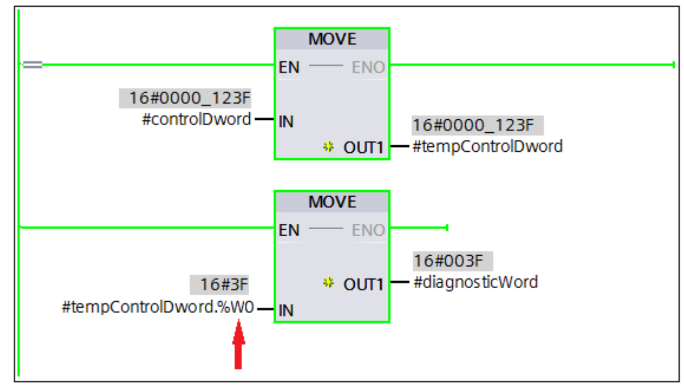
Figure 3-36:Symbolic bit, byte, word, DWord slice access



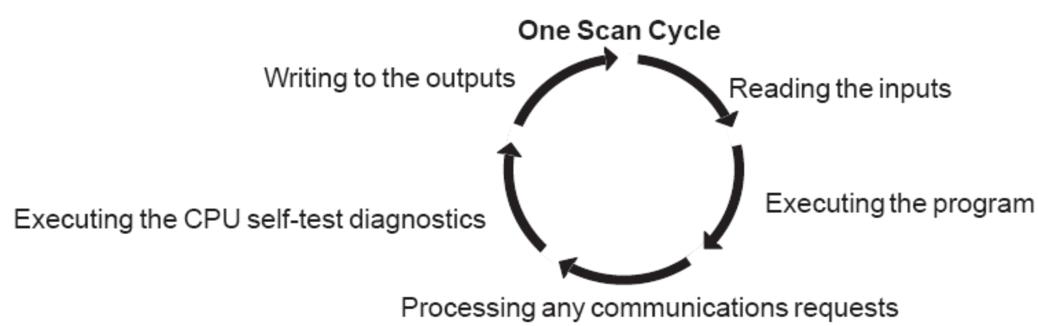






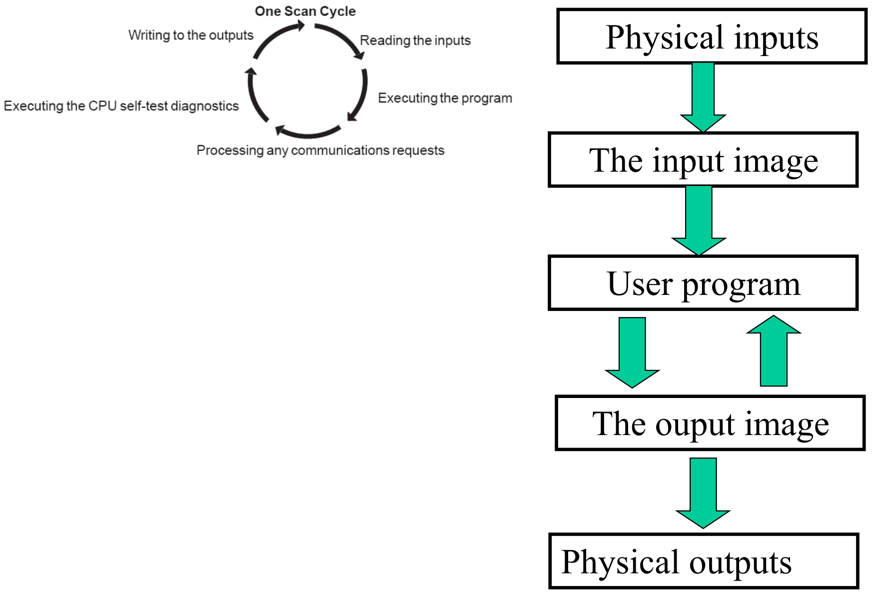


Scan cycle of PLC

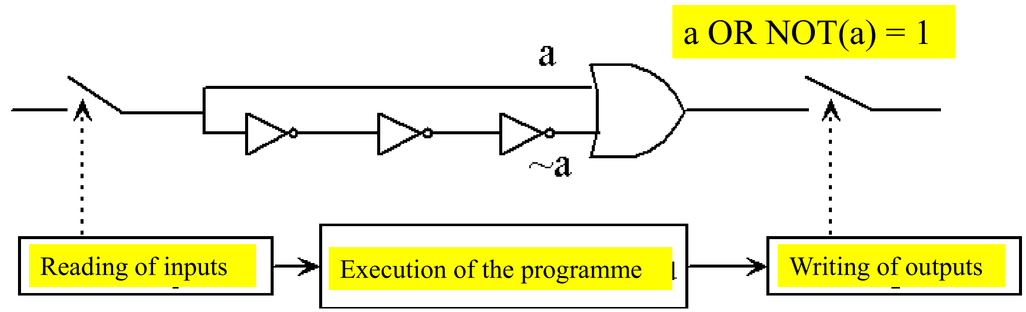


0.1 - 300 ms

Scan cycle of PLC



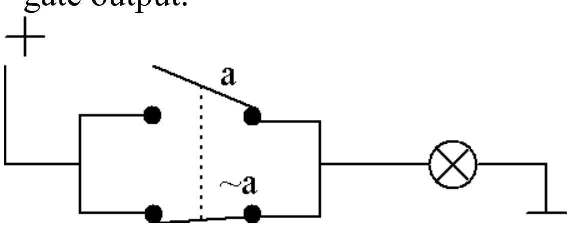
The synchronous scan cycle eliminates hazard

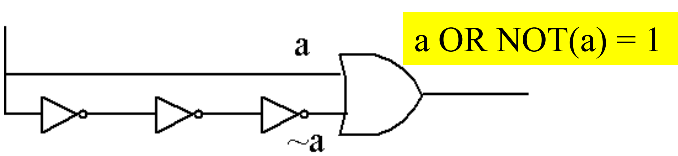


Hazard in a digital logic

In digital logic, a hazard in a system is an undesirable effect caused by either a imperfections in the system or external influences.

Example: A bulb shouldn't go out, there should always be 1 at the gate output.

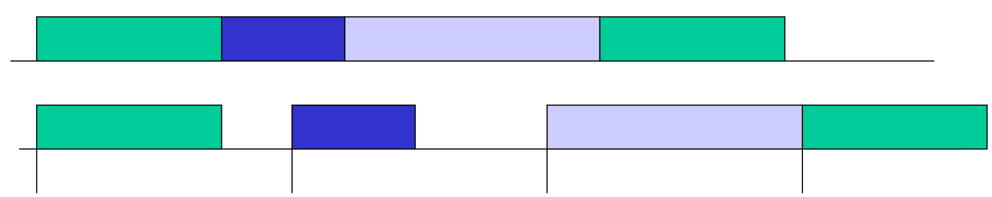


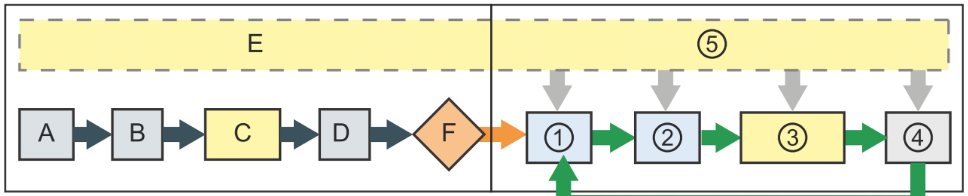


Cycle repetition rate

- Cyclic operation - successive cycles as quickly as possible
- Isochronous work - in equal intervals defined by the programmer

Cycle time	Range (ms)	Default
Maximum scan cycle time[1]	1 to 6000	150 ms
Fixed minimum scan cycle time[2]	1 to maximum scan cycle time	Disabled



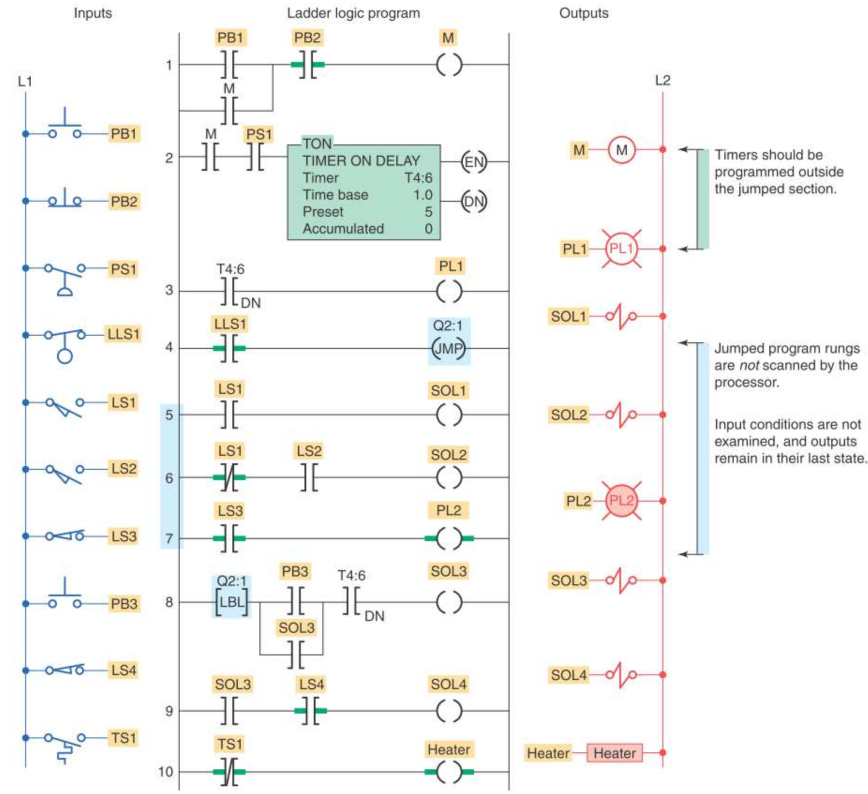


STARTUP

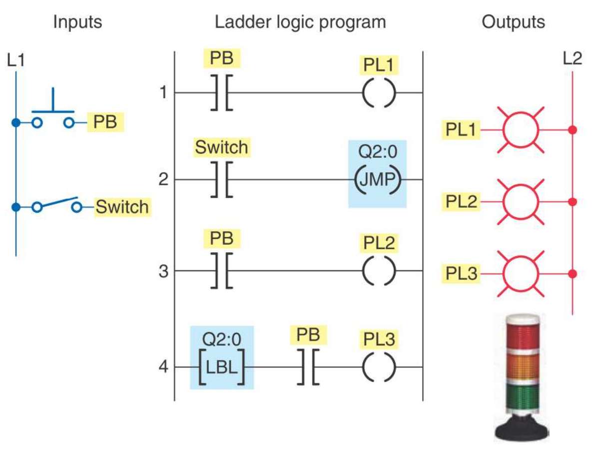
- A Clears the I (image) memory area
- B Initializes the outputs with either the last value or the substitute value
- C Executes the startup OBs
- D Copies the state of the physical inputs to I memory
- E Stores any interrupt events into the queue to be processed after entering RUN mode
- F Enables the writing of Q memory to the physical outputs

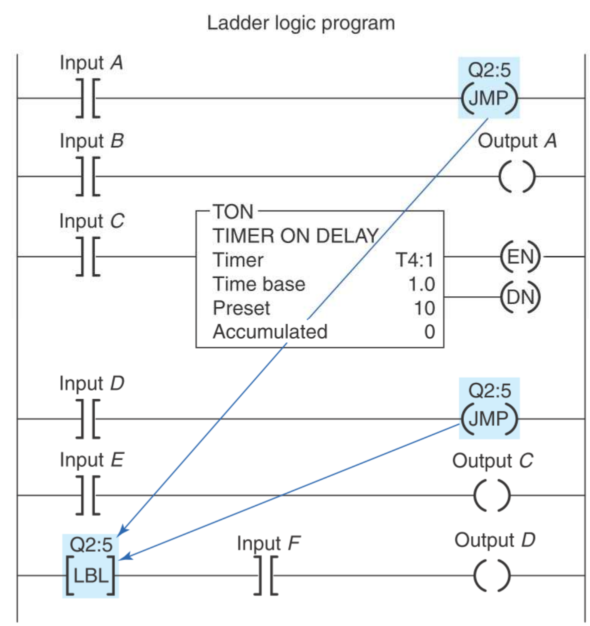
RUN

- ① Writes Q memory to the physical outputs
- ② Copies the state of the physical inputs to I memory
- ③ Executes the program cycle OBs
- ④ Performs self-test diagnostics
- ⑤ Processes interrupts and communications during any part of the scan cycle

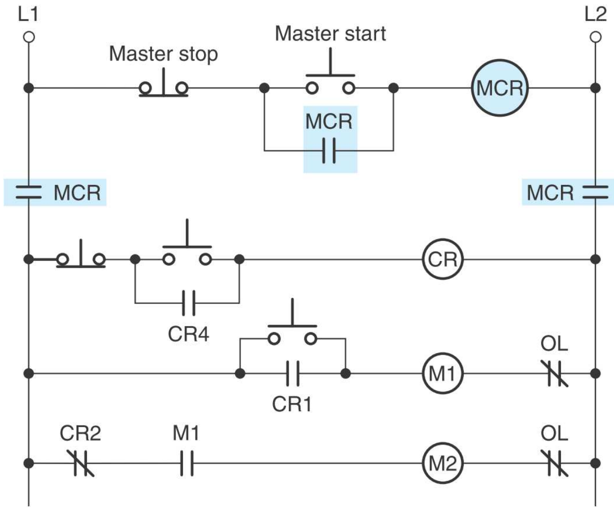


Allen-Bradley SLC 500





Hardware *Master Control Relay*



Master Control Reset (MCR)